Crossing the Membrane: Diffusion, Facilitated Diffusion and Active Transport

PhET Membrane Transport · PhET

Years 7-9	~30 minutes	Science · Biology

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Simulation: https://phet.colorado.edu/en/simulations/membrane-transport
Curriculum links: AC9S8U01 · AC9S8U02 · cell membrane · homeostasis

Learning intentions

- I can describe how solutes move across a cell membrane by diffusion.
- I can compare diffusion, facilitated diffusion and active transport.
- I can explain how the structure of the cell membrane relates to its function in keeping the cell's internal environment stable.

Getting started

Open PhET Membrane Transport. Choose a solute (such as oxygen, sodium ions or glucose) from the panel and add several particles to one side of the membrane using the pump or drag tool. Watch what happens to the particles over time before you add any transport proteins.

Tasks

1. Add a small, uncharged solute (like oxygen) to one side of the membrane with none added to the other side. Describe what happens to the particles over the next few seconds, without adding any transport proteins.

2. Now try adding a charged solute (like sodium or potassium ions) to one side only, again with no transport proteins added. What do you notice compared to Task 1?

3. Explain why the uncharged solute could cross the membrane on its own, but the charged solute could not. Use the word 'membrane' in your explanation.

4. Add a transport protein for your charged solute from the protein panel onto the membrane. What happens to the ions now?

5. Test three different solute/protein combinations. For each, record whether the particles moved from high to low concentration, or low to high.

Solute type	Transport protein added?	Direction of movement

6. Find a transport protein that moves solute against its concentration gradient (from low to high concentration). What does the simulation show is being used or required for this to happen? Explain why this type of transport is different from diffusion.

7. Set up unequal concentrations of a solute on either side of the membrane, with an appropriate transport protein in place. Let the simulation run. Describe what happens to the concentrations on each side over time.

8. Cells need to keep their internal environment stable even when the outside environment changes. Using what you've observed in this simulation, explain how transport proteins help a cell do this.

Extension (optional)

Red blood cells placed in very salty water will shrink as water leaves the cell by diffusion. Use what you have learned about concentration gradients in this simulation to explain why this happens, and predict what would happen to a red blood cell placed in pure water instead.